



الهيئة الاتحادية للهوية  
والجنسية والجمارك وأمن المنافذ  
FEDERAL AUTHORITY FOR IDENTITY,  
CITIZENSHIP, CUSTOMS & PORT SECURITY

# ID Card Toolkit Java Sample Build Instructions

Version 1.3

March 2026

CONFIDENTIAL

# Table of Contents

---

|                                                    |   |
|----------------------------------------------------|---|
| Revision History .....                             | 3 |
| Abbreviations .....                                | 4 |
| 1 Introduction .....                               | 5 |
| 1.1 Purpose .....                                  | 5 |
| 2 Java Sample Application Build Instructions ..... | 6 |
| 2.1 Prerequisites .....                            | 6 |
| 2.2 Build Instructions with Gradle .....           | 6 |
| 2.2.1 Step 1: Place SDK Libraries .....            | 6 |
| 2.2.2 Step 2: Build .....                          | 6 |
| 2.2.3 Step 3: Locate the Output .....              | 6 |
| 2.2.4 Step 4: Set Up Configuration .....           | 6 |
| 2.2.5 Step 5: Run .....                            | 7 |
| 2.3 Build Instructions with Eclipse .....          | 7 |
| 2.3.1 Step 1: Import the Project .....             | 7 |
| 2.3.2 Step 2: Place SDK Libraries .....            | 7 |
| 2.3.3 Step 3: Add JARs to Build Path .....         | 7 |
| 2.3.4 Step 4: Build .....                          | 7 |
| 2.3.5 Step 5: Set Up Configuration .....           | 7 |
| 2.3.6 Step 6: Run .....                            | 7 |
| 3 Digital Signature Service .....                  | 8 |

## Revision History

| Version | Date           | Description of Changes                                                                                   |
|---------|----------------|----------------------------------------------------------------------------------------------------------|
| 1.0     | 17th May, 2018 | Release Version 1.0.5                                                                                    |
| 1.1     | 11th Jun, 2018 | Release Version 1.0.6                                                                                    |
| 1.2     | 20th Dec, 2023 | Release Version 3.0.0                                                                                    |
| 1.3     | Mar 2026       | Toolkit v3.0.5. Updated Gradle and Eclipse instructions. Updated Digital Signature service dependencies. |

## Abbreviations

| Abbreviation | Description                                                          |
|--------------|----------------------------------------------------------------------|
| DSS          | Digital Signature Service                                            |
| ICP          | Federal Authority for Identity, Citizenship, Customs & Port Security |
| IDE          | Integrated Development Environment                                   |
| JDK          | Java Development Kit                                                 |
| SDK          | Software Development Kit                                             |

# 1. Introduction

---

The Federal Authority for Identity, Citizenship, Customs & Port Security (ICP) developed the ID Card Toolkit to address the requirements of service providers (SP) to integrate, into their business applications: Identification, Authentication, Digital Signature and Non-repudiation services, around the capabilities of the Emirates ID Card. The Toolkit is comprised of a number of Software Development Kits (SDK) supporting different programming languages and platforms.

## 1.1. Purpose

This document is a developer's reference guide to create and build the sample application provided with the ICP ID Card Toolkit Java SDK. Build instructions are provided for both Gradle (command line) and Eclipse IDE.

## 2. Java Sample Application Build Instructions

The sample project is Gradle-based and can be built with the Gradle build tool or imported into an IDE such as Eclipse or IntelliJ IDEA.

### 2.1. Prerequisites

- Java Development Kit (JDK) 1.8 or above
- ICP ID Card Toolkit Java SDK jar file (EIDAToolkit.jar)
- Gradle (latest version), or use the bundled Gradle wrapper (gradlew)
- IDE (optional): Eclipse, IntelliJ IDEA, or similar
- Windows, Linux, or macOS

### 2.2. Build Instructions with Gradle

#### 2.2.1. Step 1: Place SDK Libraries

Copy EIDAToolkit.jar (Toolkit Java SDK) and json-simple-1.1.1.jar into the libs folder of the sample project. The json-simple jar is provided with the sample project.

#### 2.2.2. Step 2: Build

Execute the Gradle build command from the project root (where build.gradle is located):

```
gradle build
```

If Gradle is not installed on the system, use the bundled wrapper:

```
./gradlew build
```

On Windows:

```
gradlew.bat build
```

#### 2.2.3. Step 3: Locate the Output

After a successful build, the executable Toolkit Sample jar file is generated in the build/libs folder of the project directory.

#### 2.2.4. Step 4: Set Up Configuration

1. Place the application configuration file config\_ap in the same directory as the generated jar file.
2. Place SP-specific configuration files in the subdirectory specified by the environment parameter in config\_ap.

For example, if config\_ap contains environment = dev, create a dev subdirectory and place the SP configuration files there.

**Note:** If `config_ap` is not provided, default configuration is used and all SP configuration files must be placed directly in the jar file directory.

### 2.2.5. Step 5: Run

```
java -jar build/libs/ToolkitSample.jar
```

## 2.3. Build Instructions with Eclipse

### 2.3.1. Step 1: Import the Project

1. Open Eclipse.
2. Go to **File > Import > General > Existing Projects into Workspace**.
3. Browse to the sample project directory and click **Finish**.

### 2.3.2. Step 2: Place SDK Libraries

Copy `EIDAToolkit.jar` and `json-simple-1.1.1.jar` into the `libs` folder of the imported project.

### 2.3.3. Step 3: Add JARs to Build Path

1. In the Package Explorer, expand the `libs` folder.
2. Right-click `EIDAToolkit.jar` > **Build Path > Add to Build Path**.
3. Repeat for `json-simple-1.1.1.jar`.

### 2.3.4. Step 4: Build

Build the project (**Project > Build Project** or `Ctrl+B`).

### 2.3.5. Step 5: Set Up Configuration

Create a folder and place `config_ap` and SP-specific configuration files as described in the Gradle instructions above.

### 2.3.6. Step 6: Run

Run the sample project from Eclipse. When prompted, provide the path to the configuration files directory.

---

## 3. Digital Signature Service

---

**Note:** AdES digital signature functionality is built into the toolkit natively. No additional JAR dependencies are required.